

BANGLADESH UNIVERSITY OF ENGINEERING AND TECHNOLOGY



Department of Electrical and Electronic Engineering

Course No : EEE 208

Group No :03

Course Title: Electronic Circuit II Laboratory

Section: A1

Project Title-

Audio Control Circuit Using Op-amp and Audio Power Amplifier

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Introduction:

Audio signal processing is an important part of electronics, used in sound systems, noise cancellation, and more. This project explored basic techniques in audio manipulation. Tasks included volume control using an LM386 amplifier and an LDR, removing specific frequencies with a band reject filter, canceling signals through destructive interference, and mixing audio with an inverting adder circuit. These tasks combined theory and practice to demonstrate key concepts in analog audio processing.

Operations:

- ✦ Volume control of an audio file (wav format) using LM386 audio amplifier and LDR.
- ✦ Cancelling out an audio file using Active Band Reject filter (**Recommended task**)
- ✦ Cancelling out an audio file using destructive interference technique or in other words algebraic operation
- ✦ Audio mixing using a inverting adder technique using ua741 op-amp.

Proposed Circuit Diagram:

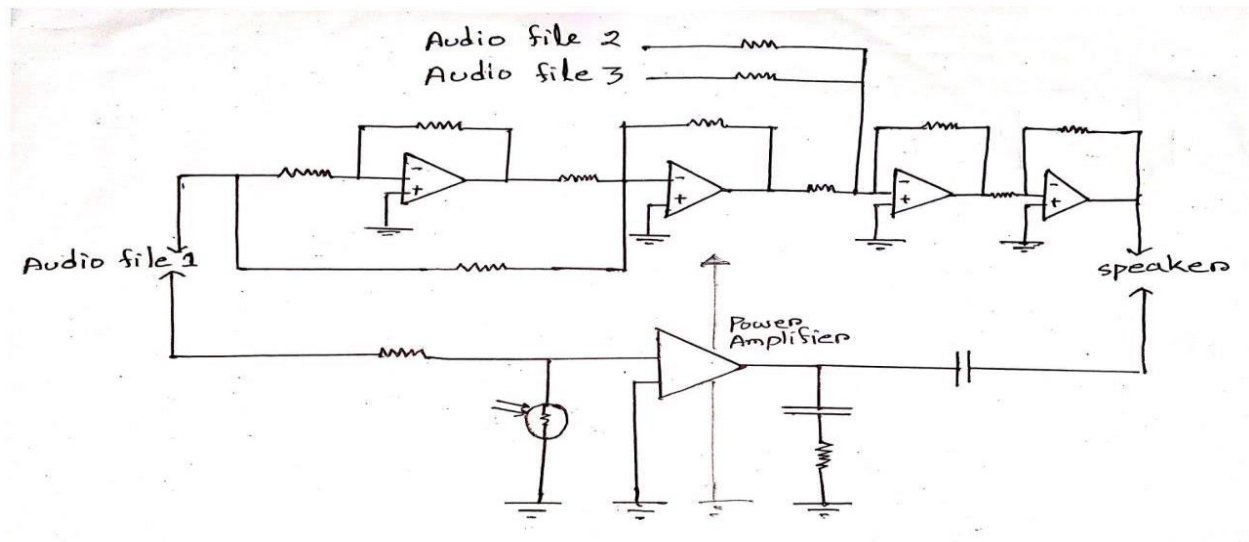


Figure 1: Proposed circuit diagram

This was our proposed circuit diagram. This circuit actually had three segment that could perform three different operation on audio file. However, we were assigned with an additional task of designing a band reject filter that would cancel out the entire audio file. The schematic diagram of both the circuits are given below:

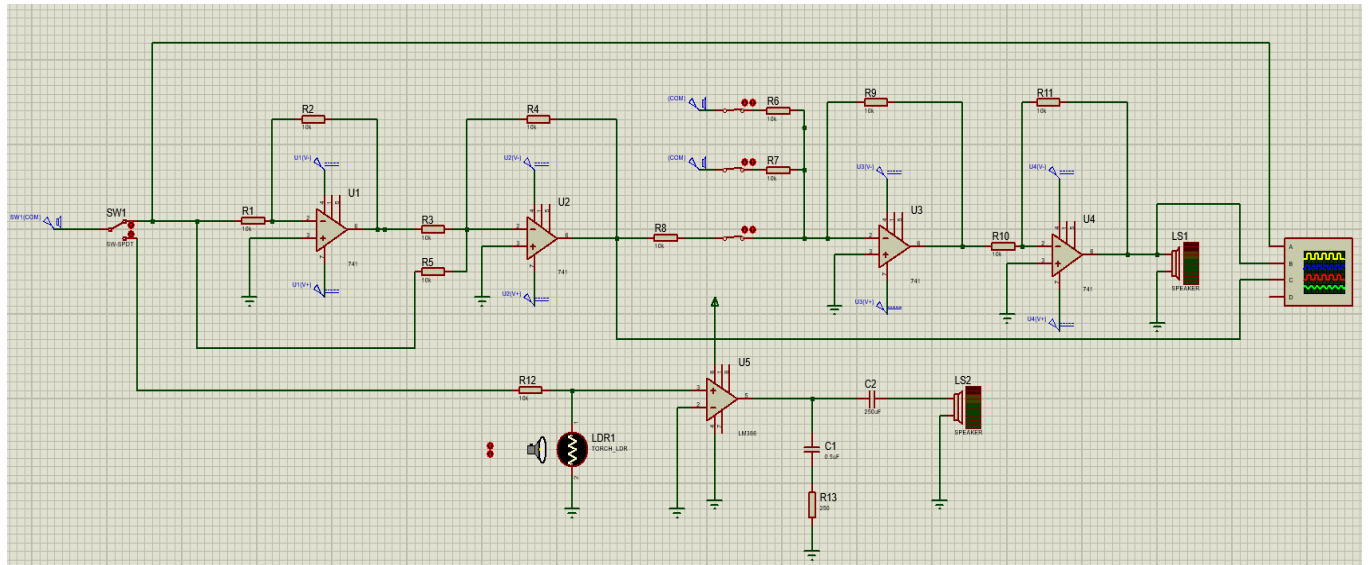


Figure 2: Schematic Diagram of circuit I

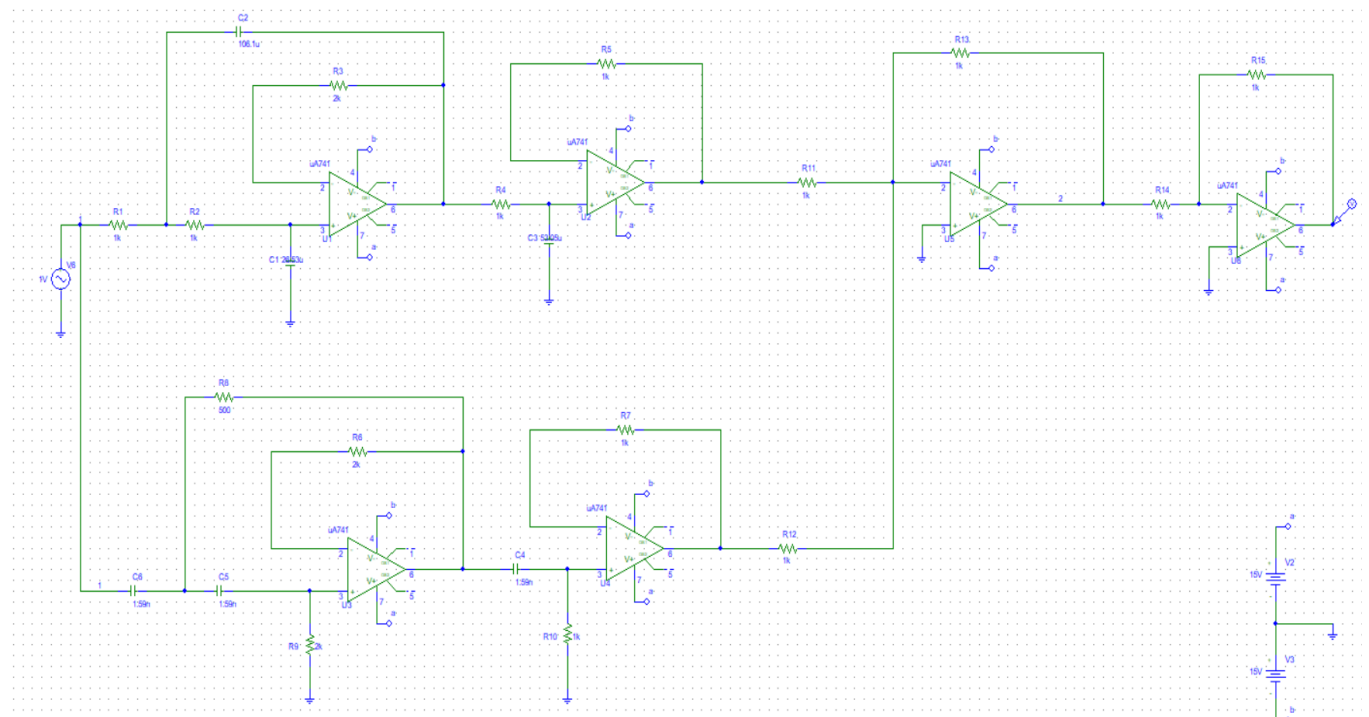


Figure 3: Schematic Diagram of circuit II

Explanations and Calculations for the operations

i) Volume control of an audio file (wav format) using LM386 audio amplifier and LDR:

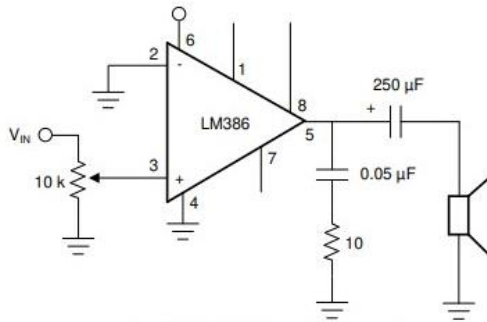


Figure 4: Circuit diagram of volume control circuit using LM 38

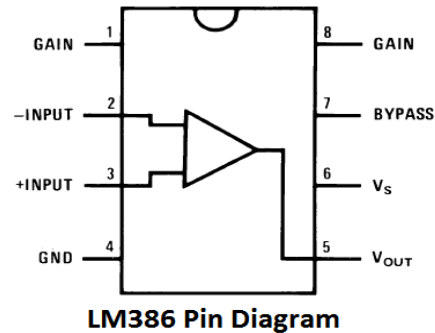


Figure 5: Pin diagram of LM386

This circuit's reference comes from Texas Instruments Incorporated's LM386 audio amplifier datasheet. The gain control PINs in this case are PINs 1 and 8. The gain is internally set to 20, but by connecting a capacitor between PINs 1 and 8, it can be raised to 200. With the right capacitor, gain may be set to any amount between 20 and 200.

The input PINs for sound signals are pins 2 and 3. Pin 2, which is grounded, is the negative input terminal. The sound signal is routed into Pin 3, the positive input terminal, to be amplified. To regulate the level in our circuit, it is connected to an audio source using an LDR. Here we have used the LDR to control the volume of the audio file. We know that the resistance of LDR decreases when we apply light source on it and increases when the light source is taken away. Hence the voltage drop across the input of the LM386 is being controlled and the amplification of the signal by 20 gain is being done which in turns control the level of volume.

Pin 4 and 6 are the power supply Pins of IC, Pin 6 for is +V_{CC} and Pin 4 is Ground. The circuit can be powered with voltage between 5-12V.

Pin 5 is the output PIN, from which we get the amplified sound signal. The output signal has both AC and DC component, and DC component is undesirable and can't be fed to Speaker. So, to remove this DC component, a capacitor C2 of 250μF has been used.

Along with this capacitor, a filter circuit of Capacitor C3 (.5uF) and resistor R1 (250) has been used at the output PIN 5. This filter also called the “Zobel network”, this electronic filter is used to remove the sudden High frequency oscillations or noise.

And finally Pin 7 is the bypass terminal. It can be left open or can be grounded using a capacitor for stability. In our circuit we have left it open.

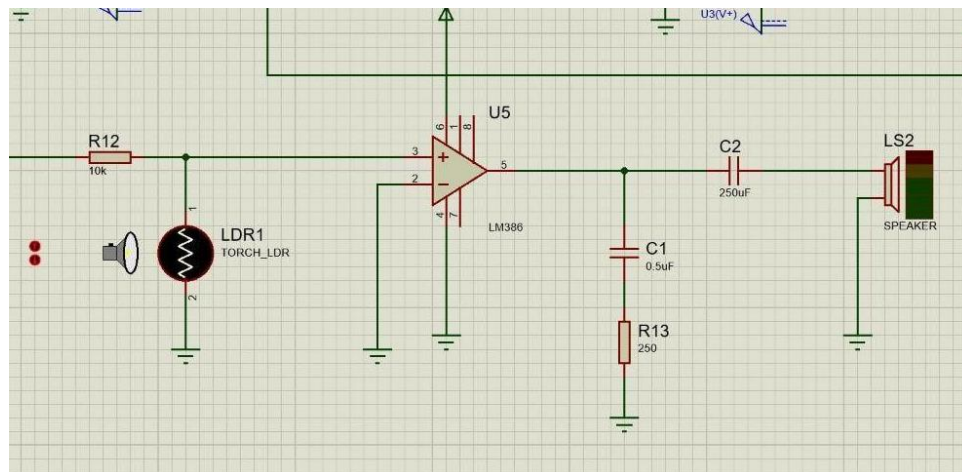


Figure 5: Schematic diagram in Proteus

ii) Cancelling out an audio file using Active Band Reject filter (**Recommended task**)

For this filter design we took reference from our text book “*Operational amplifiers and Linear Integrated Circuits*” by Robert F. Coughlin and Frederick F. Driscoll.

Here we wanted to nullify any sound between 20Hz to 20kHz. And so we tuned the circuit parameters in such way that this frequency range is in pure reject band of our output. Hence we set the cut off frequencies to be 3Hz to 100kHz to obtain the whole 20Hz to 20KHz in the stop band. The following reference was taken to design the circuit,

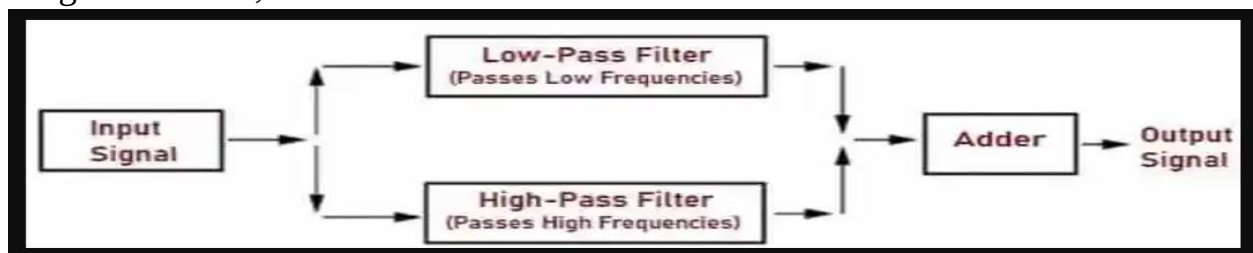


Figure 6: Block diagram of Band reject filter

In this operation we used Pspice software to design the circuit as we didn't use any real audio file. Rather we modeled the audio signal as a sinusoidal voltage signal. We used 3rd order or 60dB/decade filters to obtain the band pass filters in order to gain steep roll off. Firstly, the calculation of low pass filter is as follows:

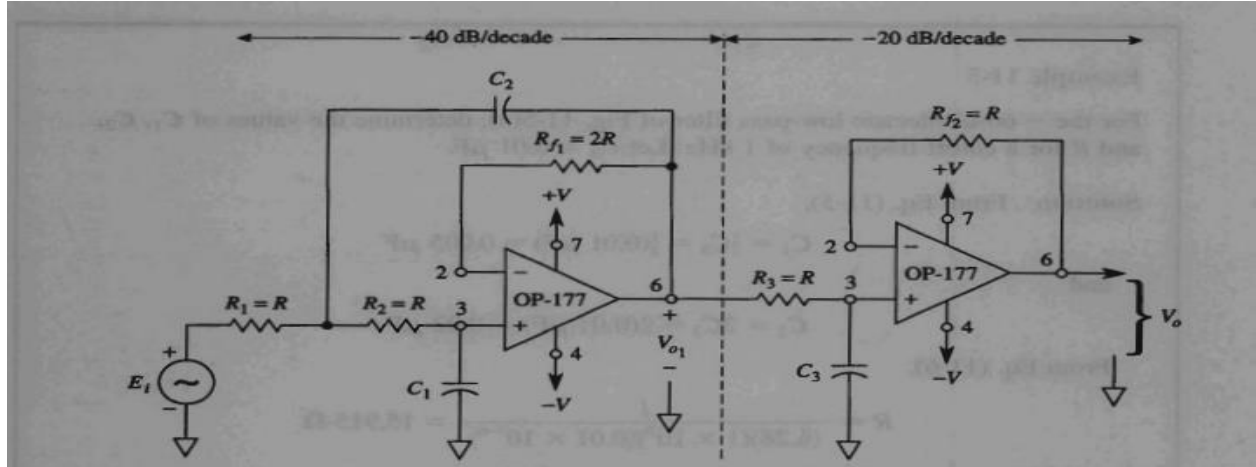


Figure 7:60 dB/decade low pass filter

And the formulae to calculate the cutoff frequency,

$$R = \frac{1}{\omega_c \times C_3}$$

$$1000 = \frac{1}{2\pi \times 3 \times C}$$

$$C = 53.05 \mu\text{F}$$

And now we will move on to the part of calculating the cut off frequencies for 3rd order 60dB/decade High pass filters,

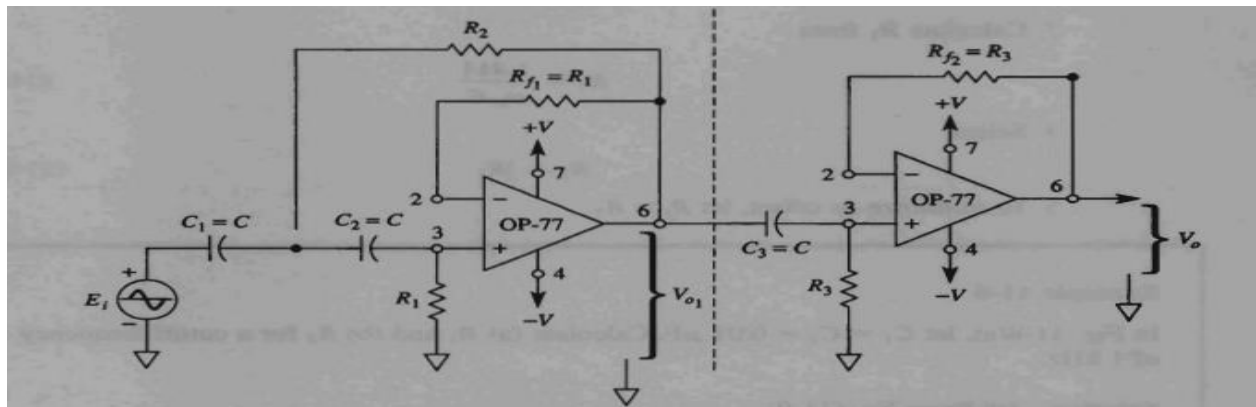


Figure 8:60 dB/decade high pass filter

And the equation to calculate the cut off frequency for this circuit,

$$R_3 = \frac{1}{\omega_c \times C}$$

$$1000 = \frac{1}{2\pi \times 100k \times C}$$

$$C = 1.59nF$$

And then we added these two filters through an inverting adder and inverted the final output again to restrain any phase inversion. The following is the final schematic diagram of the filter that we had designed,

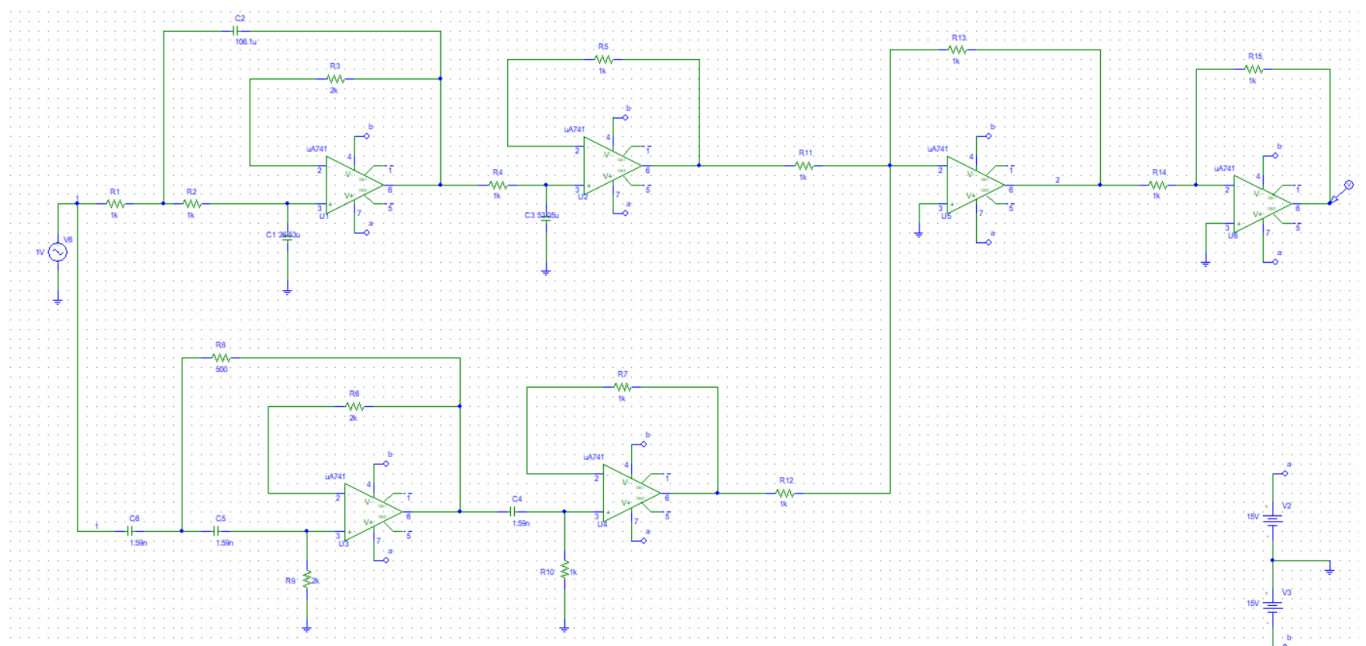


Figure 9: Band reject filter designed in Pspice

And then to observe the performance of our filter, we ran an ac sweep from 1Hz to 100kHz to see the behavior of the filter. And the frequency response of the filter was as follows:

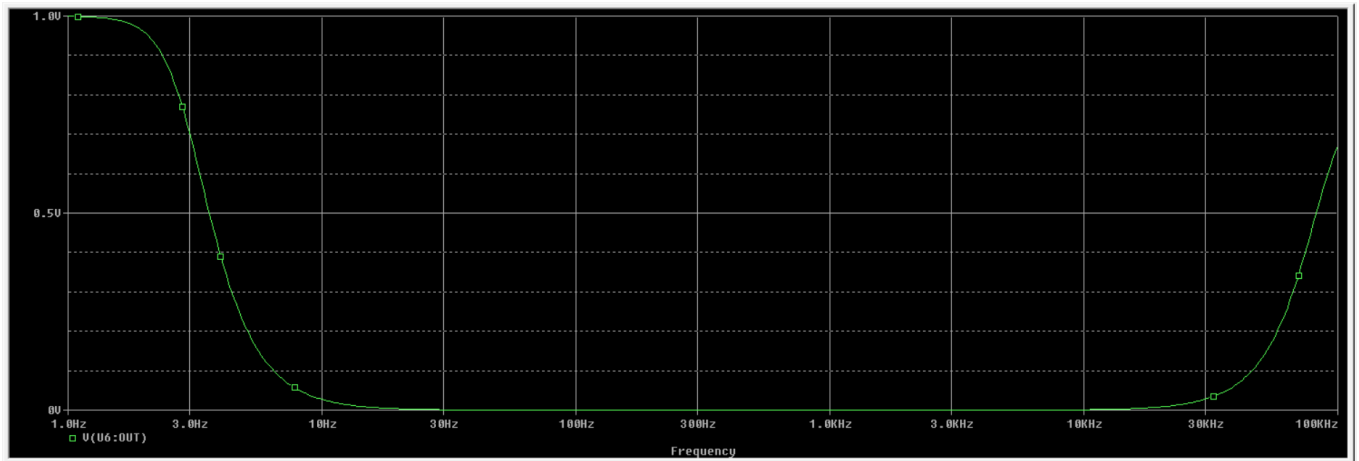


Figure 10: Frequency response of our designed filter

iii) Cancelling out an audio file using destructive interference technique or in other words algebraic operation:

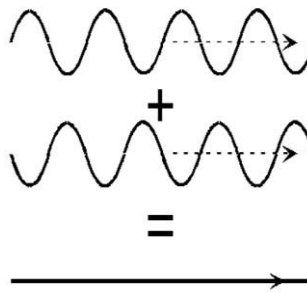


Figure 11: Nullifying through destructive interference

Active noise cancellation (ANC) is a technology that uses sound waves to reduce unwanted noise. It works on the principle of destructive interference: two sound waves of equal amplitude and opposite phase cancel each other out when they meet. While the concept is straightforward, practical implementation of ANC faces several challenges:

1. **Phase Matching:** The inverted noise signal must be precisely in phase with the original noise to achieve perfect cancellation. Any phase mismatch reduces the effectiveness of the system.
2. **Delay:** Delays in the signal processing chain can degrade the performance of the system.

In our case we first took an noise signal and then inverted its phase through an inverting amplifier and then superimposed it on the original signal. The schematic diagram of this portion of the circuit is given below,

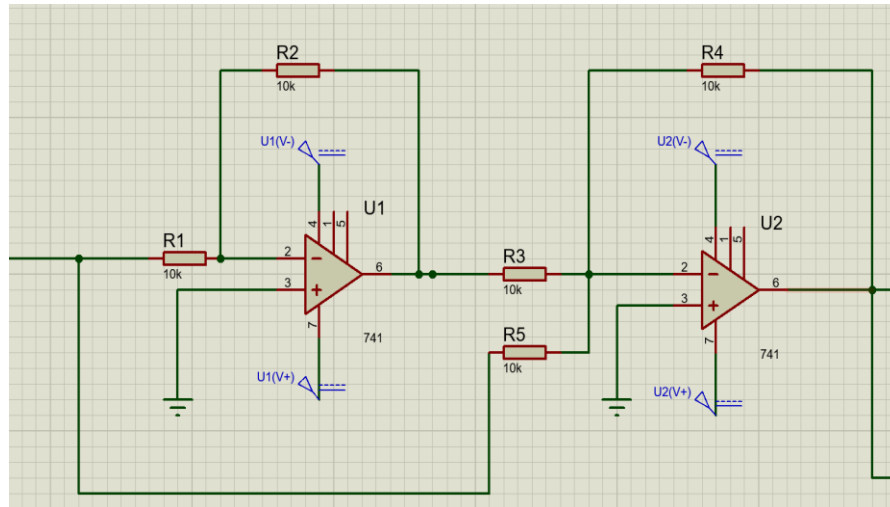


Figure 12: Audio nullifying using destructive interference technique

To observe the output of this portion, we have used digital oscilloscope of proteus design suite. Here the yellow output is the audio signal of the original file and red output is the nullified signal,

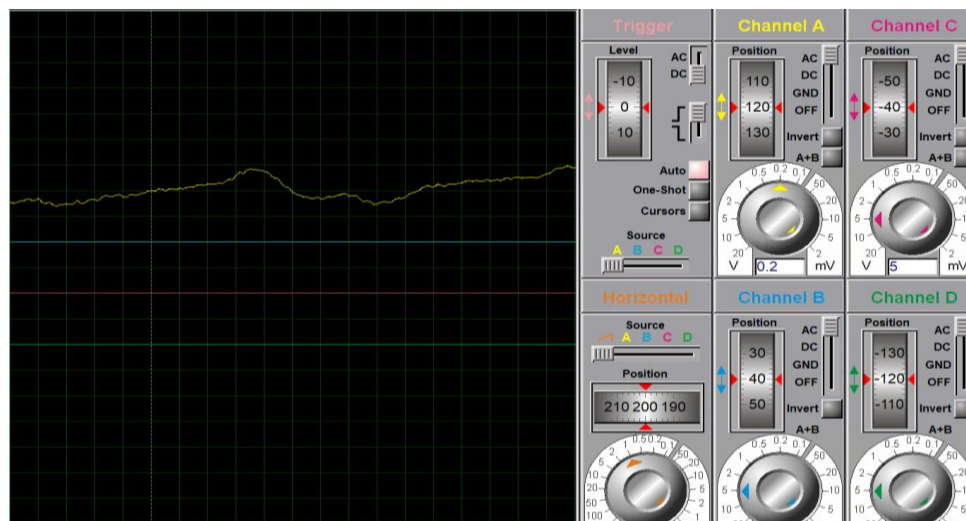


Figure 13: Visualization of destructive interference of audio signal

Here we can observe that the audio file is getting nullified properly through our destructive interference technique.

iv) Audio mixing using an inverting adder technique using ua741 op-amp.

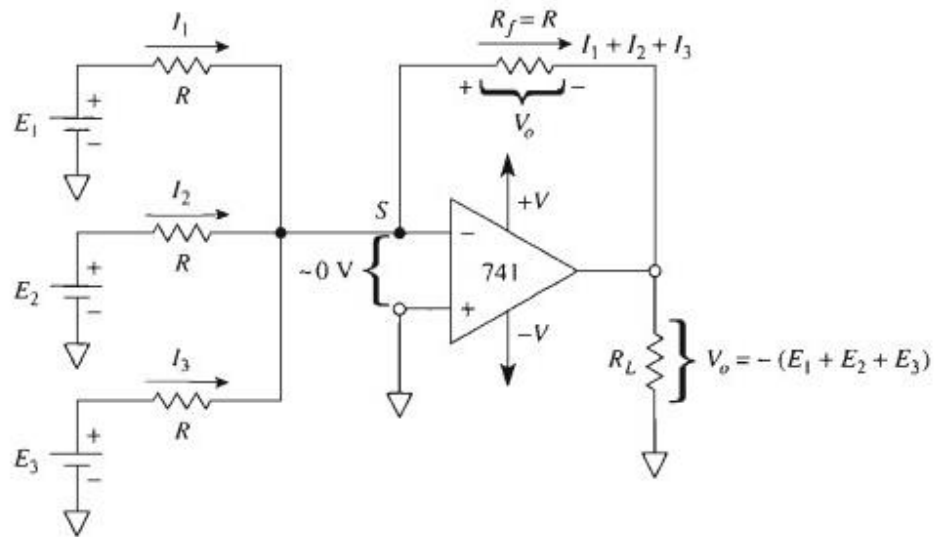


Figure 14: Audio mixing using an inverting summer technique

Multiple input signals can be combined using an adder circuit, also known as a summing amplifier, as long as the input currents are independent and do not conflict with one another. By keeping the summing node at virtual ground potential, where each input "sees" ground and only affects its own current, this is accomplished. This function is very useful in audio mixers because it allows microphone input sources, such as guitar or voice, to be merged without distortion or inter-channel crosstalk. The relative amplitudes of the signals can be changed by adding volume controls (such as 100 k Ω potentiometers) to each input. This allows for the amplification and balancing of softer signals, like a weak vocalist's voice, versus louder ones, like guitar music. This architecture guarantees versatility, clear output, and accurate mixing.

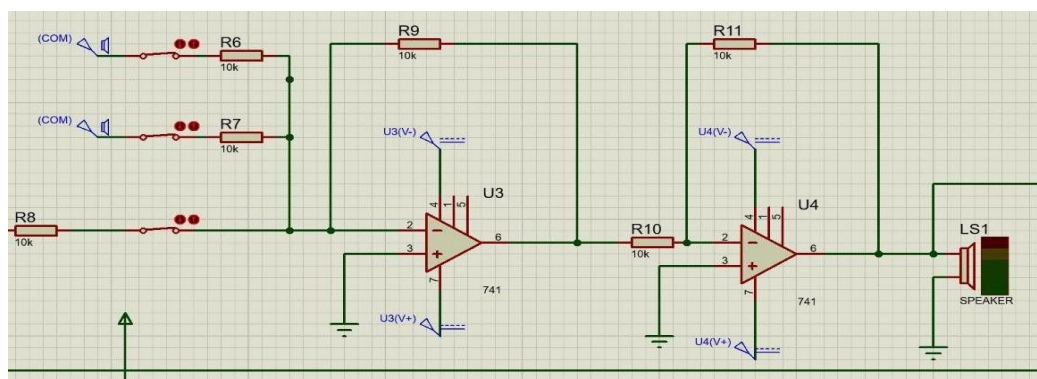


Figure 15: Schematic diagram of audio mixer in Proteus

Application:

The techniques explored in this project have several real-world applications:

- **Volume Control:**
 - Used in audio devices like speakers, headphones, and amplifiers for user-adjustable sound levels.
 - LDR-based systems can enable light-controlled audio systems in interactive installations.
- **Frequency Cancellation (Band Reject Filter):**
 - Noise cancellation in communication systems.
 - Eliminating specific unwanted frequencies in audio production or tuning systems.
- **Destructive Interference:**
 - Signal cancellation in audio processing for echo removal or background noise reduction.
 - Used in phased array systems for precise signal manipulation.
- **Audio Mixing (Inverting Adder):**
 - Combining audio signals in mixing consoles for music production and broadcasting. Multi-channel audio systems in theaters and live performances.

These applications highlight the practical utility of analog audio processing techniques across consumer electronics, communication systems, and professional audio systems.

Limitation:

The project had a few limitations, including a limited frequency range due to the specific components used. For example, the ua-741 op-amp that we used gives proper response in frequency range up to 100k. Hence the frequency response of our filters gets altered after 100k frequency. Also, the ua-741 op-amp has slew rate limitations and hence we can see that in the mixing technique audio files are getting slightly distorted. Also, susceptibility to noise interference in analog circuits, imprecise filter performance due to component tolerances, challenges in signal alignment for destructive interference, scalability issues for more complex systems, and reliance on external light conditions for the LDR-based volume control, affecting consistency.

Discussion:

The project provided practical insights into analog audio signal processing. The **volume control** task demonstrated dynamic adjustment using an LM386 and LDR, though reliance on external light reduced consistency. The **band reject filter** effectively removed frequencies but faced precision challenges due to component tolerances. **Destructive interference** illustrated signal cancellation but required precise alignment, limiting practical use. The **audio mixing** task showed signal combination via an inverting adder but highlighted analog limitations like noise and bandwidth constraints. Overall, the project demonstrated key techniques but suggests future improvements through digital tools and precision components.